

The intestinal microbiota as a key modulator of acute graft-versus-host disease

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Abstract

Although allogeneic haematopoietic cell transplantation (allo-HCT) is a curative therapy for various malignant diseases, severe complications such as graft-versus-host disease (GVHD) limit its use. The intestinal microbiome has long been known to modulate allo-HCT outcomes. Studies in the past two decades alone have uncovered a complex interplay between the microbial repertoire and the host immune system during allo-HCT. Preclinical studies have characterized the crosstalk between the microbiome and the immune response of the host, discovered associations between microbial taxa and the integrity of the mucosal intestinal barrier, and investigated the role of microbial metabolites in GVHD. Clinical studies have demonstrated that dysbiosis is an independent predictor of both transplantation-related and GVHD-related mortality, and ongoing trials are investigating microbiota-focused approaches to improve clinical outcomes and reduce GVHD severity after allo-HCT, paving the way for therapeutic applications of microbiome research. We anticipate that these insights will support the development of personalized therapies for patients receiving allo-HCT, integrating microbiome profiles with individual risk data. In this Review, we summarize current preclinical and clinical studies, providing a comprehensive account of translational efforts in this highly dynamic field.

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Introduction

Allogeneic haematopoietic cell transplantation (allo-HCT) is a life-saving treatment for patients with haematological malignancies, which replaces a diseased bone marrow with healthy cells from a genetically similar, but not identical, donor. A conditioning regimen administered before the haematopoietic cell infusion consisting of cytotoxic chemotherapy, sometimes combined with total body irradiation or monoclonal antibodies, eradicates pathogenic cells. This process allows for the engraftment of donor haematopoietic cells, which rebuild the immune system by restoring haematopoiesis, offering a potential cure. Additionally, donor T cells elicit a graft-versus-tumour effect by targeting any remaining disease-causing cells. However, allo-HCT is often complicated by graft-versus-host disease (GVHD), in which alloreactive donor T cells attack the host's organs, primarily targeting the gastrointestinal tract, liver and skin (Fig. 1).

Acute GVHD (aGVHD) is graded from I through to IV (mild to very severe), depending on the clinical presentation and organ-specific laboratory findings^{1–3}. GVHD prophylaxis is an integral part of allo-HCT and needs to strike a balance between inhibiting donor T cells and maintaining the graft-versus-tumour effect. Depending on transplant conditions (for example, the conditioning regimen, transplant source

and degree of immunological mismatch) and institutional practice, approaches can vary but typically consist of pharmacological T cell suppression via a calcineurin inhibitor (that is, cyclosporine or tacrolimus) combined with an antimetabolite (that is, mycophenolate mofetil or methotrexate). In addition, T cell depletion can be achieved in the graft prior to transplantation or afterwards using anti-thymocyte globulin or post-transplant cyclophosphamide⁴. Despite these efforts, grade II–IV aGVHD has a 6-month cumulative incidence of 35–46%⁵ and remains a major cause of mortality⁶. Grade I aGVHD is often untreated, with the exception of grade I skin aGVHD which can be treated with topical steroids. In other organs, the typical first-line treatment for grade II or higher aGVHD is systemic steroids⁴ or sirolimus, an immunosuppressive mTOR inhibitor sometimes used off-label to avoid the use of steroids, and which exhibits non-inferiority from the steroid prednisone for standard-risk aGVHD⁷. Approximately 50% of patients do not respond to first-line therapy⁸ and receive second-line treatments such as the Janus kinase 2 (JAK2) inhibitor ruxolitinib⁴. However, a substantial number of patients do not achieve a durable response even to second-line therapy, emphasizing the unmet need for novel treatment approaches^{9,10}.

Early studies of GVHD pathology discovered the underlying driving mechanism to be an excessive activation of donor T cells in response

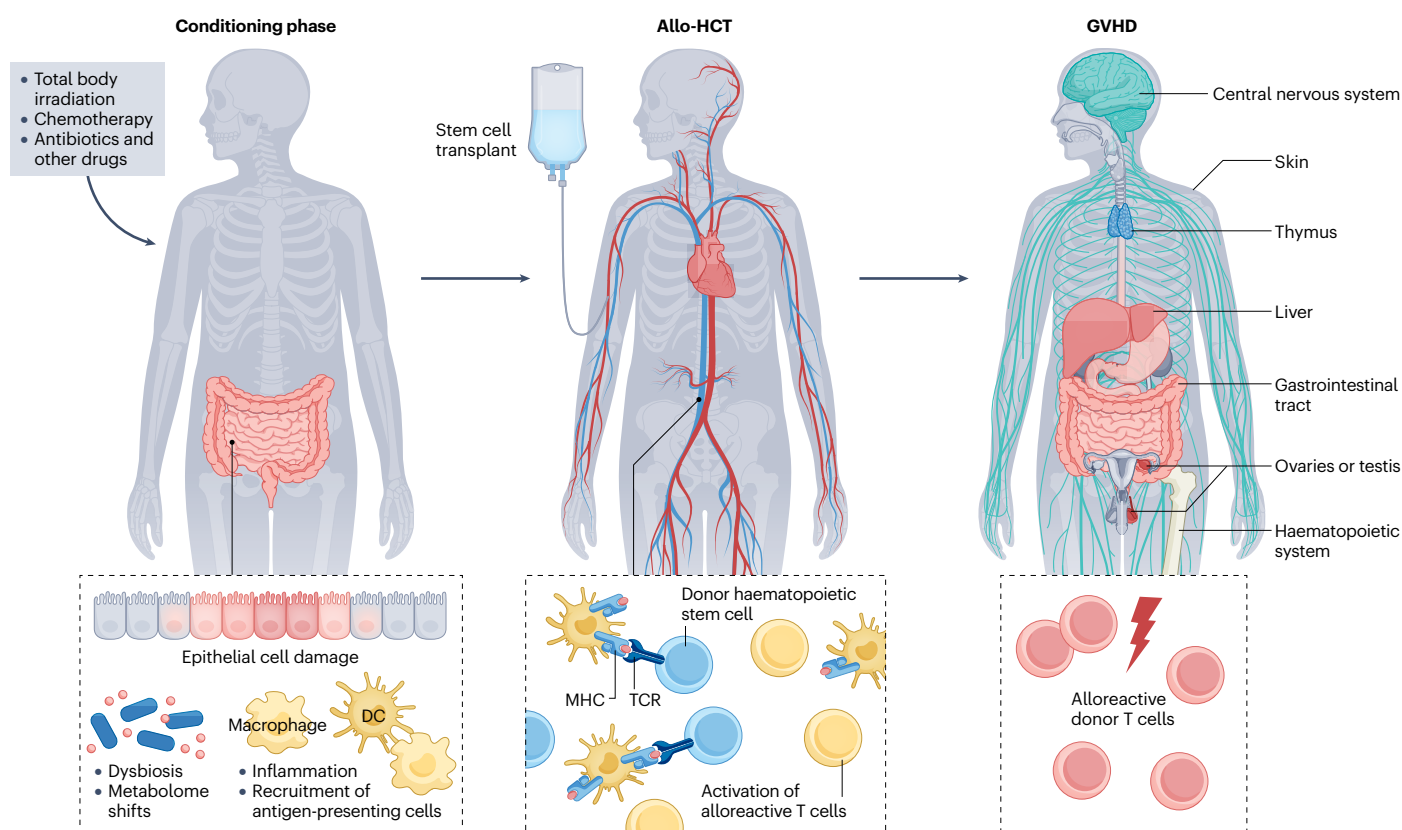


Fig. 1 | Allogeneic haematopoietic cell transplantation and graft-versus-host disease. Patients undergoing allogeneic haematopoietic cell transplantation (allo-HCT) are treated with a pre-transplant conditioning regimen consisting of chemotherapy, with or without total body irradiation and/or antibody-based therapies, to eradicate malignant cells and to allow for the engraftment of donor haematopoietic cells. Several other drugs are routinely given to combat side effects of the treatment, such as pain medication and antiemetics. Antibiotics are administered prophylactically to prevent and to treat infections. All of these

drugs are potent inducers of microbial dysbiosis and shifts in the bacterial metabolome. Following this regimen, donor haematopoietic cells are infused into the patient. Graft-versus-host disease (GVHD) arises when alloreactive donor T cells target and damage host tissues, including the skin, liver, gastrointestinal tract, central nervous system, thymus, reproductive organs (ovaries or testes) and haematopoietic system. DC, dendritic cell; MHC, major histocompatibility complex; TCR, T cell receptor.

Box 1 | The role of the virome and mycobiome in graft-versus-host disease

In addition to bacteria, the human gut harbours a wide variety of viruses and fungi, many of which remain unexplored^{188,189}. Recent studies indicate that the virome and mycobiome interact with the bacterial community to influence host immunity and overall physiology^{190,191}. Little is known regarding their impact on allogeneic haematopoietic cell transplantation (allo-HCT) outcomes, but some studies point to potential roles in shaping graft-versus-host disease (GVHD) pathology and faecal microbiota transplantation (FMT) efficacy.

The gut virome is primarily composed of bacteriophages — viruses that propagate via infection of bacteria — which vastly outnumber other viral types in both diversity and abundance¹⁸⁹. By contrast, eukaryotic viruses, which infect human cells, typically represent a minor fraction of the virome under healthy conditions but can become more prominent in disease or immunocompromised states. One study longitudinally evaluated the virome of 44 patients undergoing allo-HCT, finding a significant increase in the proportion of vertebrate viral sequences following transplantation¹⁹². In this same study, a positive association between levels of the eukaryotic *Picobirnaviridae* and acute gastrointestinal GVHD was found¹⁹², although these viruses may now be classified as bacteriophages¹⁹³. Bacteriophages can modulate bacterial metabolism by imparting metabolite-encoding genes on their hosts, altering the gut metabolome for their own benefit and affecting the mammalian host's physiology¹⁹⁴. How these shifts impact clinical outcomes after allo-HCT is just beginning to be explored: Thiele Orberg et al.⁴⁰ longitudinally studied the intestinal microbiome, mycobiome and virome along with targeted metabolic profiling in 78 patients undergoing allo-HCT. The study uncovered a microbial signature of Lachnospiraceae and Oscillospiraceae and associated bacteriophages correlating with immunomodulatory metabolites (IMMs). A risk score based on IMM abundance was developed, associating low IMM levels with higher risks for reduced overall survival, GVHD and relapse. In patients classified as low risk for these complications by the score, short-chain fatty acid (SCFA) synthesis pathways, and specifically the butyric acid-synthesis gene *BCoAT*, were enriched. The authors identified two bacteriophages encoding

the gene, providing intriguing evidence for a clinical benefit derived from bacteriophage-mediated modulation of gut bacteria⁴⁰. Another recent study analysed a multi-omics dataset consisting of bacteriome, virome, metabolome and drug intake data from patients receiving allo-HCT and found correlations between metabolic pathways, bacteriophage species and bacterial taxa, which in turn associated with relapse or remission³⁶. Overall, the impact of the virome on the microbiome composition and metabolic function during allo-HCT needs to be better understood, but these findings thus far suggest underappreciated interactions and may provide the basis for bacteriophage therapy to target specific bacterial taxa. In patients with *Clostridioides difficile* infections treated with FMT, the transfer of *Caudovirales* bacteriophages has been associated with positive treatment outcomes¹⁹⁵.

Even less is known to date about the role of resident gut fungi during allo-HCT. Allo-HCT likely impacts the fungal community, with two studies involving 78 patients and 156 patients obtaining measurements of stable density and diversity, but the latter study reported changes in species composition upon detailed longitudinal analysis^{40,196}. Arguing that mycobiota dysbiosis may not be reflected in summary metrics of bacterial composition obtained from sequencing analyses, the researchers of this latter study also performed culture-based studies, finding that faecal samples collected post transplant were more likely to yield culturable fungi compared with pre-transplant samples, and identifying patients with fungal dysbiosis based on culture positivity and expansion of *Candida parapsilosis* complex species. These patients had lower overall survival and higher transplant-related mortality¹⁹⁶. Another culture-based study of 153 patients undergoing allo-HCT found that oral or gastrointestinal colonization with *Candida* species was positively associated with the development of acute GVHD and, specifically, gastrointestinal GVHD¹⁹⁷.

Together, these findings suggest that the composition of the virome and mycobiome is altered during collapse of the gut ecosystem associated with allo-HCT, and that fungi and viruses may impact GVHD pathology and FMT treatment in ways that are not yet understood.

to mismatched histocompatibility of the recipient's tissue^{11,12}. Since then, factors influencing this complex immune reaction have been intensely studied, including the recipient's intestinal microbiota. During allo-HCT, severe immunocompromisation results from profound neutropenia occurring after bone marrow depletion and before engraftment of donor haematopoietic cells. Patients also receive GVHD prophylaxis, which is immunosuppressive, and thus patients become highly susceptible to infections. Therefore, to mitigate these effects patients are subsequently treated with prophylactic and often therapeutic antibiotics¹³, with detrimental effects on the intestinal microbiome. The intestinal microbiome comprises a community of microorganisms, which is dominated by bacteria but also includes viruses and fungi. However, although our current understanding of the impact of the gut mycobiome and virome on allo-HCT outcomes is limited (Box 1), the role of gut bacteria in maintaining immune balance, supporting the intestinal barrier and producing immunoregulatory metabolites such as short-chain fatty acids (SCFAs) and secondary

bile acids (SBAs) has been comprehensively studied in recent years. Moreover, dysbiosis, which is the disruption of the function and composition of the microbiome characterized by loss of microbial diversity and expansion of individual taxa, has been associated with allo-HCT outcomes, including GVHD^{14–24}.

Since our previous review in 2018 (ref. 25), mechanistic studies in mouse models have elucidated complex interactions between the gut microbiome and the immune system in GVHD, including how altered microbial communities influence immune regulation, epithelial barrier integrity and metabolite-mediated signalling pathways. Several clinical trials have explored microbiome interventions, including randomized controlled trials with the potential to demonstrate a cause-and-effect relationship between the composition and diversity of the gut microbiome and clinical outcomes. Although some trials have yielded unexpected or null results^{26,27}, these studies have refined our understanding and will guide further study design towards translating findings from controlled animal models to the dynamic post-transplant human setting.

In this Review, we provide an updated synthesis of the literature to explore progress made in our mechanistic understanding of dysbiosis-mediated GVHD and in therapeutic translation of these findings. We also provide evidence supporting the role that personalized microbiome-based interventions could have in revolutionizing GVHD management.

The intestinal microbiota and allo-HCT

Overall survival following allo-HCT is linked to microbial diversity

Before and during allo-HCT, chemotherapy, changes in diet, inflammatory processes and exposure to drugs such as antibiotics can lead to intestinal dysbiosis^{15,28,29}. In addition, GVHD can induce intestinal inflammation, in turn affecting microbiota composition³⁰. Several single-centre analyses have found that microbial dysbiosis is negatively associated with overall survival after allo-HCT^{15,17,20,31}.

In a landmark study in 2020, Peled et al.²⁴ found that gut microbial diversity predicted mortality after allo-HCT. In this study, 16S ribosomal RNA sequencing analysis of faecal samples from a large cohort of patients at four centres in the USA, Germany and Japan revealed loss of microbiota diversity during transplantation, commonly including domination by bacterial taxa such as *Enterococcus* and *Klebsiella*. Dysbiosis patterns were comparable across geographical locations. Patients with higher versus lower gut diversity, particularly during the neutrophil peri-engraftment period (days 7–21 post transplant), had a 29–51% lower risk of mortality. Those patients receiving broad-spectrum antibiotics such as the penicillin and β -lactamase inhibitor combination piperacillin–tazobactam and the carbapenem drug meropenem for the treatment of infectious complications had lower microbiota diversity and increased mortality risk. Similarly, those patients with lower microbiota diversity had a higher risk of transplantation-related and GVHD-related mortality, but not a higher risk of relapse, suggesting that microbial diversity largely influences immune-mediated complications rather than tumour control. Interestingly, patients in the study exhibited microbiota disruption even before transplantation when comparing the composition of their baseline faecal sample with that of healthy volunteers. Although higher microbial diversity before allo-HCT was associated with lower risk of death in only one of the four cohorts, the authors suggest that this time point could present an early opportunity for microbiota restoration strategies. Furthermore, the study identified a predictive signature of microbiota (that is, *Enterococcus* dominance) associated with a higher risk of mortality.

The gut microbiome is associated with key adverse outcomes after allo-HCT

Besides overall survival, the gut microbiome has been linked to other adverse outcomes of allo-HCT, such as neutropenic fever, relapse and aGVHD and chronic GVHD (cGVHD). Up to 80% of patients undergoing allo-HCT experience at least one episode of neutropenic fever during the prolonged neutropenic phase induced by cytotoxic chemotherapy and/or total body radiation³². Neutropenic fever after allo-HCT has been positively associated with an increased abundance of *Akkermansia muciniphila*³³, mirroring findings in patients with acute leukaemia who had a higher risk for neutropenic fever following *Akkermansia* expansion³⁴. *A. muciniphila* breaks down mucin, the main component of the intestinal mucus layer, as its primary nutrient source. In mouse models of neutropenic fever, the expansion of *A. muciniphila* leads to thinning of the colonic mucus layer and increased abundance of inflammatory cytokines, which can act as endogenous pyrogens. The effect

can be reversed by targeted antibiotic eradication of *A. muciniphila* or by administration of the SCFA propionate, which likely inhibits mucin utilization by the bacteria³³.

Relapse of the primary malignancy remains the leading cause of death after allo-HCT³⁵. Known associations with the microbiota include a higher risk of relapse with low abundance of *Blautia*¹⁸ and likelier remission with high abundance of *Eubacterium limosum*²¹. A recent study also linked various microbial enterotypes with remission status (see further discussion below)³⁶.

As early as the 1970s, a link between the microbiome and aGVHD pathology was established, with studies in germ-free mice and antibiotic-treated transplant mouse models suggesting that gut bacteria have a key role^{37,38}. Several microbiome signatures and specific bacterial taxa have since been linked to both the incidence and the severity of aGVHD in humans (Fig. 2). For instance, Burgos da Silva et al.³⁹ found that recipients of allo-HCT with lower-gastrointestinal aGVHD had worse survival outcomes and reduced levels of strict anaerobic bacteria, particularly members of the class Clostridia and genera that produce the SCFA butyrate, such as *Blautia*, *Anaerostipes*, *Eubacterium* and *Ruminococcus*. Following the onset of aGVHD, butyrate-producing bacteria continue to decline, whereas facultative anaerobes such as *Enterococcus* expand. Patients with lower-gastrointestinal aGVHD also exhibit decreased microbial pathways for SCFA fermentation, particularly butyrate production, and increased pathways for antibiotic resistance and nucleotide biosynthesis as demonstrated by functional metagenomic analysis. These shifts reflect broader metabolic changes resulting from the dysbiotic state of the gut. Furthermore, the enrichment of antibiotic resistance genes suggests selection for resistant strains following prior antibiotic exposure, contributing to persistent dysbiosis and limiting the effectiveness of standard antimicrobial therapies. In line with other analyses⁴⁰, the noted increased abundance of microbial nucleotide biosynthesis pathways likely reflects a shift towards rapidly proliferating, inflammation-associated bacteria that dominate the gut ecosystem in aGVHD³⁹. Another study similarly linked reduced levels of *Blautia* and Ruminococcaceae, as well as low levels of SCFAs, with severe aGVHD⁴¹. More recently, Hayase et al.⁴² reported a lower abundance of *Bacteroides ovatus* in patients with steroid-refractory lower-gastrointestinal aGVHD and showed in a mouse model that reintroducing this bacterial species led to reduced GVHD severity and improved survival.

The connection between dysbiosis and cGVHD is a more recent focus of investigation, with growing evidence supporting its relevance. A small study (11 patients with cGVHD and 13 patients without GVHD) found a greater loss of alpha diversity after allo-HCT in patients who went on to develop cGVHD. In addition, butyrate-producing *Clostridium* was present in a higher abundance at pre-transplant baseline in patients with cGVHD and showed a significantly greater decline post engraftment compared with controls⁴³. Another study found a positive association between the lower circulation of SCFAs such as butyrate at 3 months post transplant and the development of cGVHD following allo-HCT⁴⁴. The oral microbiome is also gaining attention in cGVHD studies. A longitudinal analysis of 29 patients with cGVHD versus 51 individuals without GVHD revealed an oral expansion of *Streptococcus salivarius* and *Veillonella parvula* and a high degree of overlap between oral and gut bacterial species, suggesting ectopic gut colonization⁴⁵. Another recent study found that oral dysbiosis induced by mucositis secondary to conditioning therapy was associated with increased cGVHD severity. In a mouse model, bilateral molar ligatures (an orthodontic technique) in recipients of allo-HCT disrupted the oral flora,

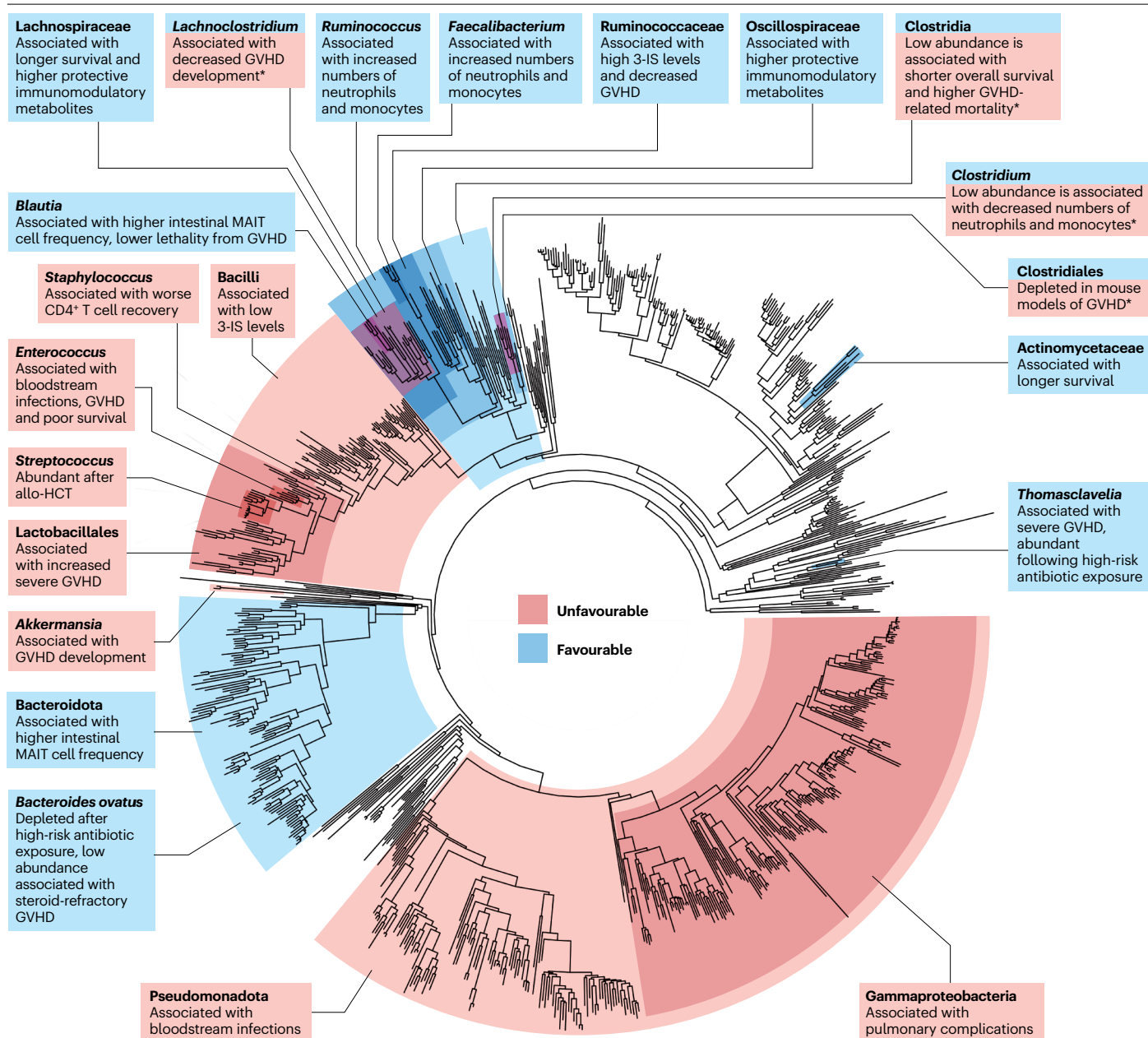


Fig. 2 | The impact of the microbiome on allogeneic haematopoietic cell transplantation outcomes. Phylogenetic tree illustrating the common gut bacterial species, with shaded clades representing different taxonomic classifications. Bacteria are associated with favourable effects (blue clades) or unfavourable outcomes (red clades) following allogeneic haematopoietic cell

transplantation (allo-HCT). *Beneficial commensals that when present in lower relative abundance have been associated with unfavourable outcomes. The 20 publications that encompass this figure were published between 2010 and 2024 (refs. 15–21,24,31,39,40,42,44,66,127,181–185). 3-IS, 3-indoxyl sulfate; GVHD, graft-versus-host disease; MAIT cell, mucosal-associated invariant T cell.

exacerbated cGVHD and led to an expansion of Enterococcaceae in the oral cavity and the gut, also indicative of ectopic gut colonization⁴¹. Habenicht et al.⁴⁷ recently explored whether dysbiosis has a role in cGVHD pathogenesis via B cell dysregulation, finding that patients who went on to develop cGVHD already showed an increase in CD20⁺CD38^{hi} plasmablasts by day 90 after allo-HCT. These plasmablasts generated antibodies targeting specific commensal bacteria, particularly those

producing SCFAs, suggesting that dysregulated mucosal immune responses to gut microbiota may contribute to cGVHD pathogenesis.

Overall, these findings suggest that gut microbiota diversity and composition before transplantation and during the peri-engraftment period is a robust predictor of treatment outcomes in patients undergoing allo-HCT, independent of geographical location and clinical practices.

Allo-HCT-associated factors modulate the gut microbiome and affect clinical outcomes

During allo-HCT treatment, patients receive many medications that influence both the composition and the function of the intestinal microbiome. Antibiotics are administered to prevent and control infections during allo-HCT. However, antibiotics increase the risk for major adverse outcomes, including, but not limited to, GVHD. For instance, the anaerobe-targeting antibiotic metronidazole has been positively associated with intestinal domination by *Enterococcus*, which is a risk factor for vancomycin-resistant *Enterococcus* (VRE) bacteraemia¹⁵. Early administration of azithromycin to prevent bronchiolitis obliterans syndrome after allo-HCT has been positively associated with *Bacteroides fragilis* abundance and risk of tumour relapse^{36,48}. Finally, treatment with broad-spectrum antibiotics such as β -lactams and carbapenems has been consistently and reproducibly associated with increased incidence of intestinal aGVHD and steroid-refractory GVHD^{19,23,42,49,50}. The respective gastrointestinal environment often shows an expansion of *Bacteroides* and low concentrations of butyrate and propionate in the colonic lumen^{42,51–53}.

Intestinal microbial diversity declines after conditioning treatment, particularly in patients receiving myeloablative regimens rather than non-myeloablative regimens⁵⁴. Total body irradiation, which is sometimes used in conditioning, has also been found to reduce microbial diversity in mouse models^{33,55}. Nguyen et al.⁵⁶ recently developed PARADIGM (parameters associated with dynamics of gut microbiota), an algorithm that integrates faecal microbiome profiles and daily medication records. In addition to identifying the expected negative association between broad-spectrum antibiotics and microbial diversity, the model also revealed several non-antibiotic drugs that adversely affect microbial diversity. Antiemetics and opioids, which are frequently administered to respectively counteract chemotherapy-induced nausea and pain due to mucositis, were associated with increased *Enterococcus* abundance and decreased gut microbial diversity. Whether these medications directly impact the microbiome, or whether the microbiome is perturbed by reductions in oral food intake that occurs as a result of nausea and oral pain, is an open question. Importantly, however, the observed microbiota shifts correlated with poorer treatment outcomes, including higher rates of GVHD and infection-related complications in independent validation cohorts. This study also demonstrated a link between antibiotic use and increased genetic convergence within dominant bacterial strains; that is, different strains became genetically more similar over time, indicating reduced competition between subspecies that could result in the selection of resistant and potentially pathogenic bacteria. Together, these findings highlight the importance of carefully managing pharmacological exposures in patients receiving allo-HCT to preserve microbiota diversity and reduce the risk of immune-mediated complications.

Patients undergoing allo-HCT often experience mucositis, loss of appetite, nausea and vomiting, all of which negatively impact nutritional status⁵⁷. Many patients therefore receive parenteral nutrition or enteral nutrition. Enteral nutrition is increasingly linked to better clinical outcomes, and changes in microbial composition may mediate these effects⁵⁸. For example, in a recent randomized study, patients receiving primarily enteral nutrition compared with parenteral nutrition had higher abundance of taxa associated with protection from GVHD, such as *Ruminococcus bromii* and *Faecalibacterium prausnitzii*⁵⁹. By contrast, long-term use of total parenteral nutrition (≥ 10 days) was associated with a loss of *Blautia*, whereas increased levels of *Blautia* were found to be associated with reduced GVHD-related mortality¹⁸.

Furthermore, dietary components such as fibre have been linked to changes in the gut microbiota that influence outcomes of allo-HCT. For example, in mice, a fibre-free 'Western' diet leads to decreased microbiota diversity and increased *A. muciniphila* abundance^{60–62}. In humans, a high-fibre diet has been linked to higher bifidobacteria abundance and faecal butyrate concentrations⁶³, which are both positively associated with favourable outcomes following allo-HCT^{15,64}. By contrast, a diet high in fermented foods has been reported to have a positive association with microbiota diversity and a negative association with inflammatory markers⁶⁵. A multicentre study found that dietary lactose promoted the expansion of *Enterococcus*, which can metabolize lactose and thrives in a lactose-rich environment, contributing to increased gut permeability and inflammation. Correspondingly, in mouse models of allo-HCT, dietary interventions such as lactose restriction have been found to reduce the expansion of *Enterococcus* and alleviate GVHD severity⁶⁶. In a recent clinical study, patients undergoing allo-HCT and following a lactose-free diet had significantly reduced colonization with VRE⁶⁷. Together, these findings highlight how diet shapes gut microbial composition and suggest that nutritional strategies could mitigate GVHD.

Mechanisms of dysbiosis-mediated GVHD

Microbial dysbiosis shapes the development and progression of GVHD by exacerbating gut permeability and creating a permissive environment for inflammation. These effects are mediated by microbiota-driven pathways that regulate host immune responses, including through modulating major histocompatibility complex (MHC) class II molecule expression and T cell clonal selection. The intestinal metabolome is also a key mediator of GVHD, with microbial and host-derived metabolites influencing allogeneic immune cell reactivity. Figure 3 summarizes the interplay between the gut microbiota and the host immune system at the intestinal barrier.

The cause and consequence relationship between intestinal damage and dysbiosis during GVHD

Several recent studies have explored the interplay between allo-HCT-related dysbiosis, intestinal barrier integrity, immune responses and GVHD-related outcomes. Microbiome-mediated degradation of mucus in the colon is a key factor contributing to damage to the intestinal barrier in GVHD. Hayase et al.⁵¹ found that administration of meropenem during -10 to $+30$ days relative to allo-HCT positively associates with acute intestinal GVHD incidence in patients, and that meropenem treatment leads to a dysbiotic state with an expansion of *Bacteroides*. This finding was recapitulated in a mouse model: meropenem-treated mice had aggravated intestinal GVHD, severe epithelial damage and a thinner intestinal mucus layer compared with controls. Moreover, these mice exhibited an expansion of *Bacteroides thetaiotaomicron* in the gut, decreased levels of luminal xylose and upregulated expression of glycoside hydrolases from *B. thetaiotaomicron*, which are enzymes responsible for breaking down mucin glycoproteins in the intestinal mucus layer. Oral supplementation of xylose improved GVHD-related survival and prevented thinning of the colonic mucus layer, suggesting that xylose suppresses the mucus-degrading behaviour of *B. thetaiotaomicron*⁵¹. In this same model, the introduction of *B. ovatus* reduced GVHD-related mortality. However, this treatment made no difference in meropenem-untreated mice, suggesting that *B. ovatus* only ameliorates GVHD when the gut is in a dysbiotic state. This beneficial effect was also only observed when *B. ovatus* was able to degrade xylose-containing polysaccharides and produce xylose, thus improving

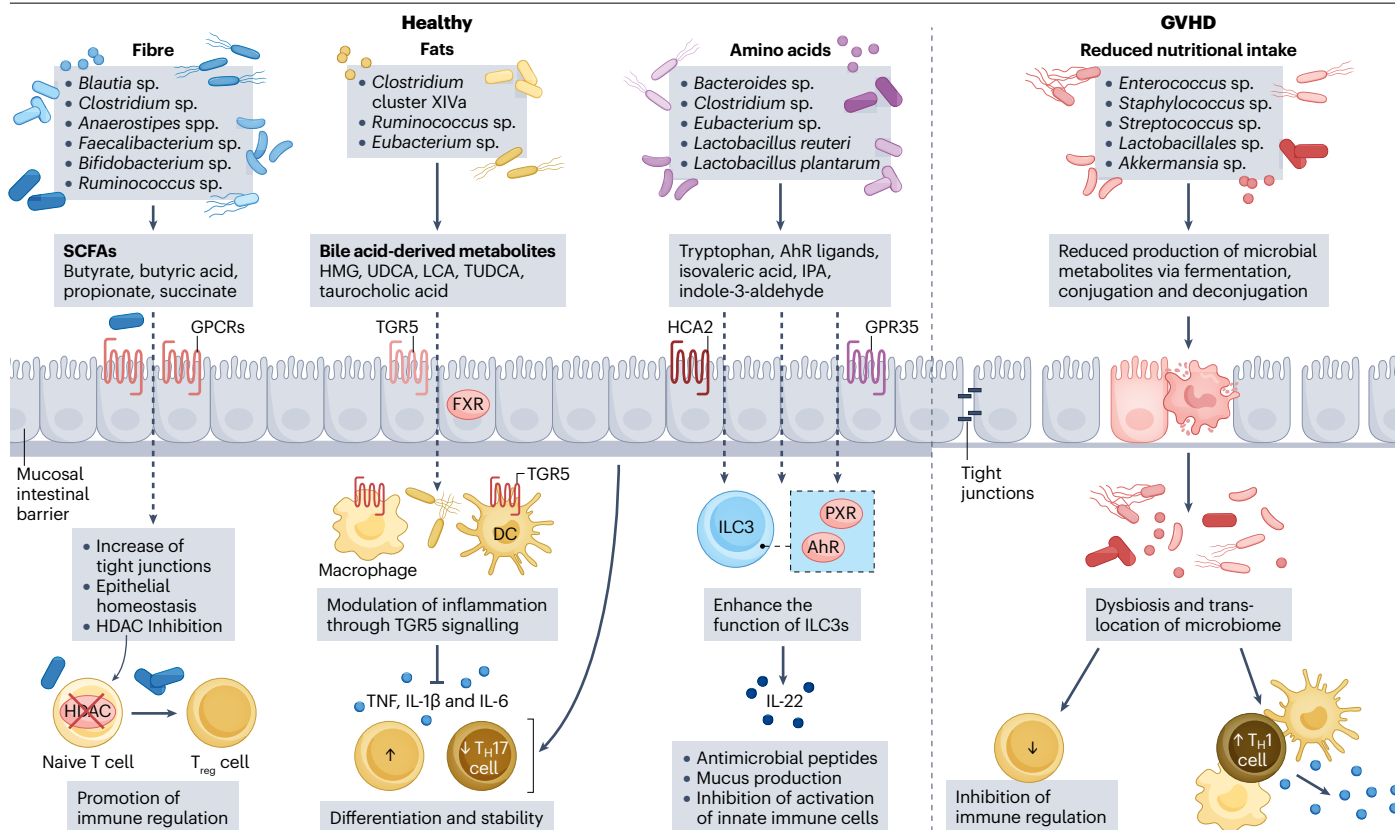


Fig. 3 | Microbiome-derived metabolites and immune interactions at the intestinal barrier in health versus graft-versus-host disease. Representation of the intestinal barrier in health and during graft-versus-host disease (GVHD), highlighting how dietary components shape microbial metabolites that in turn regulate epithelial barrier integrity and immune homeostasis. In healthy states, dietary fibre supports the growth of microbial taxa such as *Blautia*, *Clostridium* and *Faecalibacterium*, which ferment fibre into short-chain fatty acids (SCFAs) including butyrate, propionate and succinate. These metabolites increase the protein production of epithelial tight junctions and immune tolerance via G-protein-coupled receptor (GPCR) signalling and inhibition of histone deacetylases (HDACs). Dietary fats contribute to the generation of bile acid derivatives including ursodeoxycholic acid (UDCA), lithocholic acid (LCA), tauroursodeoxycholic acid (TUDCA) and hydroxymethylglutarate (HMG), through microbial metabolism performed by bacterial species such as *Clostridium* cluster XIVa and *Ruminococcus*. These metabolites activate the farnesoid X receptor (FXR) and Takeda G-protein-coupled receptor 5 (TGR5), found on epithelial cells, immune cells (such as macrophages) and enteric neurons. Activation of these receptors modulates epithelial responses by enhancing barrier integrity, stimulating antimicrobial peptide secretion and

regulating bile acid transport. Additionally, TGR5 signalling limits inflammation through suppression of pro-inflammatory cytokine production and maintenance of mucosal homeostasis¹⁸⁶. Amino acids, particularly tryptophan, are metabolized by bacteria such as *Bacteroides*, *Lactobacillus reuteri* and *Eubacterium* into indole derivatives (for example, indole-3-propionic acid (IPA)) and aryl hydrocarbon receptor (AhR) ligands. These metabolites regulate the balance between T helper 17 cells (T_H17 cells) and regulatory T cells (T_{reg} cells) and enhance gut barrier function through activation of receptors including AhR, GPR35, hydroxycarboxylic acid receptor 2 (HCA2; also known as GPR109A) and pregnane X receptor (PXR). Group 3 innate lymphoid cells (ILC3s), which reside in the gut mucosa, are activated by microbial metabolites, and contribute to epithelial barrier repair and immune regulation by producing cytokines such as interleukin-22 (IL-22), which has a key role in barrier protection by inducing epithelial cell proliferation, stimulating goblet cells to secrete mucins and upregulating antimicrobial peptide expression¹⁸⁷. In GVHD, dysbiosis and impaired microbial metabolism result in reduced production of these protective metabolites, contributing to epithelial cell damage and immune dysregulation. Pro-inflammatory cytokines such as tumour necrosis factor (TNF), IL-1 β and IL-6 are upregulated, perpetuating tissue injury. DC, dendritic cell.

the intestinal metabolic environment⁴². These findings are in line with a reported positive association between exposure to the carbapenem imipenem–cilastatin and increased GVHD severity and expansion of the mucus-degrading bacterial species *A. muciniphila* as demonstrated in a mouse model of allo-HCT¹⁹. Selective microbiota-based interventions or mucin-restoring diets could therefore mitigate GVHD-associated gut damage in patients receiving allo-HCT and broad-spectrum antibiotics.

A study by Seike et al.⁶⁸ suggests that dysbiosis is a consequence rather than a cause of GVHD. These investigators found no pathogenic

effects of allo-HCT-induced dysbiosis in naive mice and investigated whether colonic oxygen levels instead have a role in GVHD. Their findings suggest that suppressing excess oxygen levels alleviates GVHD through both microbiome-dependent and microbiome-independent mechanisms. They first demonstrated that co-housing of untransplanted, healthy mice with mice that had undergone allo-HCT and showed dysbiosis and GVHD, or gavage of untransplanted, healthy mice with the stool of these dysbiotic mice, did not result in gastrointestinal damage, weight loss or immune alterations in the absence of allo-HCT,

despite a reduction in microbial diversity. The gut is normally a hypoxic environment, and intestinal epithelial cells (IECs) rely on microbial metabolites, such as SCFAs, as their main energy source for oxidative phosphorylation (OXPHOS)⁶⁹. However, allogenic T cells can disrupt OXPHOS in IECs⁷⁰, reducing physiological hypoxia in the intestinal lumen and, in turn, suppressing obligate anaerobes while promoting facultative anaerobes⁶⁸. In the Seike et al.⁶⁸ study, oral chelation of iron reduced intestinal ambient oxygen levels, alleviated dysbiosis and decreased the severity of gastrointestinal GVHD, but did not alter donor T cell immune responses. Intestinal oxygen levels also impacted GVHD severity via direct effects on IECs by disrupting hypoxia-inducible factor 1 α (HIF1 α) signalling⁶⁸, which crucially regulates intestinal homeostasis under low oxygen conditions⁷¹. Furthermore, IEC-specific *Hif1a*-knockout mice showed greater GVHD severity and mortality than their wild-type counterparts, despite being co-housed together to harmonize their microbiomes. This study provides critical mechanistic insights into how preserving gut hypoxia and restoring microbial balance could mitigate GVHD-associated intestinal injury.

Interplay between the intestinal microbiome and immune cell functions

The immune system is a key player in GVHD pathology via the activation of alloreactive T cells. The complex crosstalk between the gut microbiome and the host's immune system during GVHD has recently been reviewed elsewhere^{14,30,72}, and we therefore highlight only a selection of recent and noteworthy studies here.

Several groups have previously shown that in addition to haematopoietic antigen-presenting cells, non-haematopoietic antigen-presenting cells are potent inducers of GVHD^{73,74}. Stephens et al.⁷⁵ investigated the role of MHC class II expression on IECs, given that IECs form the primary barrier between the gut microbiome and the host and were likely to be involved in microbiota-specific immunity. Using conditional knockout mice lacking MHC class II expression specifically in IECs, it was shown that IEC-derived MHC class II supports gut immune homeostasis by increasing levels of microbial-bound immunoglobulin A, maintaining regulatory T cell (T_{reg} cell) populations and shaping gut microbiota composition. This study also revealed that intestinal mononuclear phagocytes can acquire MHC class II molecules from IECs, suggesting that myeloid–epithelial interactions shape the adaptive immune response in the gut. Although this study was not conducted in the context of GVHD, the findings are relevant to gastrointestinal GVHD pathology as they suggest that disruption of IEC function, for instance during damage to the intestinal barrier, may impair local immune regulation and contribute to dysbiosis and excessive T cell activation, which is characteristic of GVHD.

In line with these findings, Koyama et al.^{76,77} found that MHC class II expression by IECs in the ileum impacts the severity of GVHD, regulated in part by the microbiota. The researchers showed that germ-free mice and mice treated with broad-spectrum antibiotics exhibit reduced MHC class II expression and linked dysbiosis during allo-HCT with increased MHC class II expression on IECs, as well as increased donor CD4⁺ T cell expansion. Furthermore, mice with IEC-specific deletion of MHC class II had significantly reduced GVHD-related mortality and T cell expansion in the gut, demonstrating that MHCII⁺ IECs can induce CD4⁺-dependent lethal GVHD⁷⁶. They next showed that genetically identical mice from different vendors (harbouring different microbiota) had different expression levels of MHC class II in the ileum, which were positively correlated with CD4⁺ T cell-mediated GVHD severity when respective mice served as recipients in preclinical models of allo-HCT⁷⁷. Furthermore,

the researchers identified bacterial taxa that positively and negatively influence MHC class II expression. Accordingly, mice receiving an oral antibiotic treatment to specifically abrogate MHC class II-inducing bacteria prior to allo-HCT had attenuated GVHD severity, confirming the direct involvement of the microbiota. Finally, Koyama et al.⁷⁷ reported a similar relationship between gut microbiome composition, human leukocyte antigen (HLA) class II expression and GVHD in a cohort of patients undergoing allo-HCT. These results underscore the key regulatory role of the microbiota in modulating alloreactive immune responses during GVHD, independent of genetic factors such as donor–recipient HLA mismatch, which are traditionally viewed as the primary drivers of GVHD^{1,3}. Thus, these studies highlight the microbiota as an additional, non-genetic determinant of disease severity.

Yeh et al.⁷⁸ also challenged the prevailing view that GVHD is primarily driven by donor–recipient genetic disparity by demonstrating that recipient microbiota dictate donor T cell expansion and its role in GVHD severity. Using Bayesian modelling to analyse T cell receptor (TCR) clonality across genetically identical donor–recipient pairs, they identified a subset of donor CD4⁺ T cell clonotypes that underwent differential expansion despite identical recipient genetics. However, these selective T cell expanders were enriched and reduced, respectively, in mice with an intact microbiota versus those receiving broad-spectrum antibiotics. Thus, microbial-derived antigens drive specific T cell responses. Furthermore, microbiota-specific donor T cells exacerbated GVHD severity when recognizing microbial antigens presented by the gastrointestinal epithelium. To validate the role of the microbiota in shaping T cell responses, the researchers introduced transgenic donor T cells specific for known microbial antigens, including segmented filamentous bacteria and flagellin-expressing commensals. These T cells expanded preferentially in recipients harbouring the cognate microorganisms and augmented GVHD-mediated lethality when co-transplanted with alloreactive T cells. Moreover, this study found that microbial antigen presentation by gut epithelial cells is MHC class II-dependent, further implicating the microbiota in driving pathogenic T cell expansion.

Jarosch et al.⁷⁹ investigated the relationship between gut microbiota composition and T cell responses in GVHD. Using multimodal single-cell immune cell phenotyping of gastrointestinal biopsies combined with microbiome analyses, they found that the severity of aGVHD and exposure to antibiotics with subsequent dysbiosis were positively associated with clonal enrichment of activated CD8⁺ T cells and a relative reduction of T_{reg} cells in the gastrointestinal tract. Through analyses of differential gene expression and relative abundance of surface proteins, this study identified a reduced functionality phenotype in gut-infiltrating T_{reg} cells from patients exposed to antibiotics. Additionally, SCFA-producing bacterial taxa, including *F. prausnitzii*, were associated with increased numbers of T_{reg} cells, and these functional T_{reg} cells were able to reduce the clonal expansion of CD8⁺ T cells that are likely to drive GVHD. Finally, the researchers analysed one clinical case of steroid-refractory gastrointestinal GVHD with complete response to faecal microbiota transplantation (FMT). T_{reg} cell infiltration into the gastrointestinal tract of this patient increased after FMT, potentially indicating increased interactions between T_{reg} cells and CD8⁺ T cells. Overall, this study supports a direct role for gut dysbiosis in immune dysregulation in GVHD by promoting a pro-inflammatory immune environment that exacerbates disease severity.

Although T cells are clearly the primary inducers of GVHD, evidence from mouse models indicates that ileal neutrophils, which reach sites of inflammatory GVHD before T cells do, also interact

with resident microorganisms and impact intestinal GVHD severity. One mouse study demonstrated that ileal neutrophil infiltration post allo-HCT is microbiota-dependent and contributes to GVHD severity through the production of reactive oxygen species (ROS) and T cell activation, with germ-free conditions or neutrophil depletion significantly reducing disease severity⁸⁰. A second study traced neutrophils from the ileum to mesenteric lymph nodes, where they presented antigen via MHC class II and promoted T cell expansion. This process likely relied on bacterial translocation after allo-HCT, as antibiotic pretreatment of mice reduced the bacterial load in the ileum and abrogated neutrophil migration⁸¹. A third study found in a mouse model of allo-HCT that the administration of human β -defensin 2, an endogenous, epithelial cell-derived peptide with antimicrobial qualities, reduced neutrophil infiltration into the ileum and local T cell activation by modulating the microbiota and dampening TCR signalling, resulting in reduced GVHD severity and mortality⁸². Although corroborating studies have yet to be conducted in humans, this work highlights the influence of the microbiome on neutrophil behaviour and its downstream effects on T cell activation in aGVHD. These studies also point to neutrophil–microbiota interactions as an underexplored and potentially promising therapeutic target, alongside established T cell-focused approaches.

Vallet et al.³⁶ investigated the relationship between gut microbiota composition, immune cell profiles and relapse risk following allo-HCT. This longitudinal study of patients receiving allo-HCT revealed that distinct gut enterotypes and their associated metabolomes correlated with T cell exhaustion and relapse risk. Specifically, *B. fragilis* and related phylotypes were positively associated with relapse and an increase in the numbers of exhausted peripheral T cells, suggesting impaired antitumour immunity. Patients with an *Enterococcus*-dominant enterotype that associated with higher relapse risk showed increased primary bile acid metabolism, suggesting impaired metabolic functioning. This study also implicated the gut virome in these processes, as bacteriophages co-occurred with enterotypes associated with T cell exhaustion. These findings highlight a gut–immune axis in which microbial communities influence systemic T cell function and dysbiosis exacerbates immune exhaustion and relapse risk.

Together, these findings redefine our understanding of GVHD by identifying the microbiome as a major regulator of immune-regulated pathology. Microbiota-targeted interventions, such as modulation of gut bacterial species or inhibition of microbial antigen presentation, could therefore serve as novel therapeutic strategies to mitigate GVHD while preserving beneficial immune responses.

GVHD is associated with distinct alterations in metabolomic profiles

The metabolome encompasses all small molecules produced and used both by host cells and by the host's microbiota. Research has primarily focused on metabolites such as SCFAs, indoles and bile acids in GVHD owing to their reported efficacy in protecting the gastrointestinal tract and their potent immunomodulatory properties. Recent studies have also linked distinct alterations in metabolomic profiles to the pathogenesis of GVHD, offering potential novel therapeutic targets.

SCFAs, the primary metabolic products of dietary fibre fermentation by gut bacteria, are a crucial energy source for colonic epithelial cells (colonocytes), for which butyrate oxidation in particular provides up to 70% of the energy needs in maintaining a healthy gut barrier⁸³. SCFAs enhance the production of protective mucus by stimulating goblet cells in the intestinal epithelium to increase mucin

gene expression and, consequently, synthesis and secretion of mucus; and also upregulate tight junction proteins, strengthening the barrier against pathogens^{84,85}. Furthermore, SCFAs modulate immune cell function by promoting anti-inflammatory properties, such as by inhibiting histone deacetylases (HDACs) in T_{reg} cells to promote their differentiation and thereby suppress excessive immune responses^{86,87} (Fig. 3). The role of butyrate in GVHD has been validated in preclinical models, in which its administration mitigates both aGVHD and cGVHD⁴⁴. Therefore, targeting SCFA production, particularly that of butyrate, or administering SCFAs directly holds therapeutic potential for the management of GVHD. However, the protective effects of SCFAs may also vary with the degree of intestinal damage and the concentration of butyrate available, potentially limiting the efficacy of SCFA administration in the management of GVHD^{88,89}. For instance, Golob et al.⁸⁸ found that higher levels of butyrogenic bacteria post GVHD were associated with steroid refractoriness. This highlights the complex role of SCFAs in GVHD as this finding also aligns with studies that have shown that loss of microbial diversity, and not just butyrogenic bacteria, is associated with GVHD severity and outcomes²⁴, emphasizing the importance of personalized microbiome-based interventions.

Metabolites derived from amino acids, including indoles such as indole-3-aldehyde (I3A), also promote intestinal barrier function and prevent epithelial damage⁹⁰. Indoles are derived from tryptophan catabolism performed by the gut microbiota and can modulate immune responses by influencing the balance of the secretion between pro-inflammatory and anti-inflammatory cytokines, as well as preventing secondary infections by promoting epithelial homeostasis and barrier integrity and, therefore, preventing bacterial translocation from the large intestine⁹⁰. To maintain mucosal integrity, many indole derivatives, including I3A, engage the aryl hydrocarbon receptor (AhR) in innate lymphoid cells and other immune cells, which in turn activates the transcription of interleukin-22 (IL-22). IL-22 then promotes the production of antimicrobial peptides and the repair of damaged epithelial cells, suppressing the excessive immune responses characteristic of GVHD to prevent tissue damage⁹¹.

Bile acids facilitate the absorption of lipids in the small intestine but also impact various other biological functions, notably including the modulation of immune signalling⁹². Studies in patients with ulcerative colitis have found significantly reduced levels of the SBAs lithocholic acid (LCA) and deoxycholic acid (DCA), which in turn are also associated with decreased levels of Ruminococcaceae, a family of bacteria known to produce SBAs⁹³. SBAs activate the bile acid receptor Takeda G-protein-coupled receptor 5 (TGR5; also known as GPBAR1), leading to the suppression of pro-inflammatory cytokines and chemokines and contributing to an anti-inflammatory environment in the gut⁹⁴. Furthermore, *in vitro* studies have found that SBAs can suppress pro-inflammatory cytokine production from human peripheral blood-derived macrophages, key mediators of intestinal inflammation in irritable bowel disease⁹³. LCA specifically has been shown to impair T helper 1 cell (T_H1 cell) activation, reducing their secretion of the pro-inflammatory cytokines tumour necrosis factor (TNF) and interferon- γ (IFN γ)⁹³. It is worth mentioning that ursodeoxycholic acid (UDCA), another SBA derived from microbial metabolism that activates the TGR5 receptor, has demonstrated the capacity to regulate the pro-inflammatory responses of antigen-presenting cells and, in doing so, contribute to intestinal health and homeostasis⁹⁵. In alignment with previous research indicating that TGR5 signalling can inhibit nuclear factor- κ B (NF- κ B) activation through the cAMP–protein kinase A (PKA) pathway⁹⁶, UDCA markedly decreased the production

and release of pro-inflammatory cytokines, including TNF, IL-6, IL-12 and IL-1 β by lipopolysaccharide (LPS)-stimulated macrophages⁹⁵. However, the precise mechanism by which UDCA modulates the immune response remains unclear, as UDCA can also interact with the farnesoid X receptor (FXR), which is implicated in cholestatic diseases⁹⁷. Nonetheless, there are conflicting reports regarding the effect of UDCA on FXR activation. Some studies suggest that UDCA acts as an antagonist to FXR activation, whereas others have reported UDCA as an agonist^{98,99}.

The numerous metabolites that derive from the gut microbiota and influence GVHD interact through complex mechanisms that must be fully understood to develop targeted therapeutic strategies to mitigate GVHD. Thiele Orberg et al.⁴⁰ investigated how bacterial and bacteriophage compositions influence the production of immunomodulatory metabolites (IMMs) in patients undergoing allo-HCT. They developed the IMM risk index, based on levels of five microbial metabolites positively associated with outcome or of biological relevance for overall survival, severe GVHD, treatment-related mortality and disease relapse. Low versus high IMM risk index scores indicate high or depleted IMM levels and are associated with lower versus higher rates of GVHD-related complications, respectively. Butyrate biosynthesis pathways were enriched in patients with a low IMM risk index, in particular the butyryl-coenzyme A (CoA):acetate CoA transferase (BCoAT) pathway, which has a central role in SCFA production by gut microbiota (Box 1). Bacteriophages encoding genes for butyrate production were also identified, suggesting that the gut virome may contribute to beneficial metabolite synthesis by introducing genetic pathways associated with SCFA generation to microorganisms through transduction, a process in which bacteriophages infect host bacteria and integrate their genetic material, including genes such as those for butyrate production, into the bacterial genome (Box 1). A key finding of the study was that patients who had received broad-spectrum antibiotics during allo-HCT exhibited disruptions in bacterial and viral communities and decreased levels of SCFAs. As low levels of intestinal SCFAs have been linked to severe aGVHD⁴¹, this finding further supports the hypothesis that interventions promoting metabolite-producing bacterial consortia or supplementation with SCFA formulations could mitigate GVHD.

Another recent study analysed the serum metabolomic profiles of two cohorts of HLA-matched related donor and recipient pairs, revealing marked differences in the metabolome between patients who developed GVHD and those who remained GVHD-free¹⁰⁰. The tryptophan-derived metabolites, indoles, were higher in recipients without GVHD compared with those with GVHD, and these differences were especially evident at the onset of GVHD, suggesting that changes in metabolites might have a role in GVHD development. However, it remains to be explored whether this phenomenon contributes to the severity of aGVHD, as the imbalance in gut bacterial species observed in GVHD might be both a cause and a result of the allogeneic immune response. Overall, this suggests that both dysbiosis and allo-HCT-induced changes in host metabolism may impact GVHD pathology.

Haring et al.¹⁰¹ investigated the role of various bile acids in mitigating aGVHD while preserving graft-versus-leukaemia effects. They found that allo-HCT disrupted bile acid metabolism, reducing the intestinal pool of bile acids and increasing the expression of FXR while decreasing the expression of TGR5 in the IECs, both of which, as mentioned above, are crucial for epithelial homeostasis and immune function. They identified tauroursodeoxycholic acid (TUDCA) as the most potent

bile acid metabolite and demonstrated in vitro that it protected IECs from TNF-induced and IFN γ -induced apoptosis. In preclinical models of allo-HCT, mice treated with TUDCA had improved survival, lower histopathological GVHD scores in various target organs and lower serum cytokine levels. TUDCA also modulated antigen presentation by IECs by downregulating MHC class II expression. Despite these immunomodulatory effects, TUDCA preserved graft-versus-leukaemia activity, as donor cytotoxic T lymphocytes maintained their ability to eliminate leukaemic cells. Unlike other bile acid derivatives, and considering the direct administration of this bile acid in this experimental model, TUDCA did not significantly alter gut microbiota composition or diversity, indicating that the observed effects were independent of the gut bacteria and could instead be the result of a direct interaction with the epithelium. Overall, this study highlights the translational potential of TUDCA as a therapeutic for preventing GVHD-related intestinal injury that may offer protection against gut inflammation while maintaining anti-leukaemia immunity.

Lindner et al.¹⁰² expanded our understanding of the role of microorganism-derived SBAs in regulating immune responses during GVHD. Using both mouse models of allo-HCT and patient samples, they demonstrated pronounced losses of microorganism-derived metabolites during GVHD. Furthermore, mice exhibiting GVHD had significantly reduced abundance of the gene encoding microbial bile salt hydrolase (BSH), the protein product of which helps transform primary bile acids into secondary ones. Metagenomic analysis of faecal samples from patients with GVHD revealed a loss of key bacterial taxa responsible for bile acid metabolism. In line with the mouse data, functional metagenomic analysis confirmed the downregulation of the gene encoding BSH, and uncovered the downregulation of genes such as *baiA*, *baiB*, *baiCD* and *baiE*, which catalyse α -dehydroxylation, another key reaction in SBA production. These reductions in specific bacterial enzymes could lead to a decrease in the overall pool of SBAs and their binding to host cell receptors such as FXR and TGR5. By reducing the signalling through these anti-inflammatory receptors, the loss of microorganism-derived SBAs contributes directly to the immune responses and intestinal damage observed in GVHD.

Lindner et al.¹⁰² also tested in vitro how several bile acids affect FXR signalling, finding that the host-derived primary bile acid chenodeoxycholic acid (CDCA) induced strong FXR signalling, whereas various microorganism-derived SBAs lowered CDCA-induced FXR activation, suggesting that microbial bile acid metabolism impacts FXR activation. In accordance with this, T cell-specific FXR ablation reduced GVHD histopathology scores in the large intestine and liver, but not in other organs, suggesting that FXR signalling may modulate allogeneic T cell functions in organs exposed to microorganism-derived bile acids. Furthermore, in mice with GVHD, pharmacological activation of FXR led to increased mortality, whereas genetic ablation of FXR in donor T cells decreased mortality.

Finally, Lindner et al.¹⁰² also revealed associations between the purported FXR antagonist UDCA and both reduced T cell proliferation and lower risk of GVHD-related mortality in patients, indicating the therapeutic potential of inhibiting FXR activity to restore microbial bile acid metabolism and mitigate GVHD. Possible approaches would include replenishing bacterial strains capable of bile acid metabolism, such as *Clostridium* cluster XIVa and *Ruminococcus* species, or supplementation with specific SBAs such as UDCA. Lindner et al.¹⁰² showed that supplementation with UDCA in patients with GVHD restored bile acid balance and limited T cell-driven inflammation, in part, by inhibiting FXR activation via its antagonistic activity as well

as through counteracting the pro-inflammatory effects of primary bile acid accumulation.

Han et al.¹⁰³ also investigated the interplay between the gut microbiota, bile acid metabolism and immune responses. Using single-cell RNA sequencing, they found significant differences in immune cell signatures between patients with and without GVHD, as well as genes related to microbiome-associated immune regulation. Moreover, in the GVHD group, a dysregulated IL-1-mediated immune response was identified, characterized by increased expression of IL-1 β , IL-1 receptor type 1 (IL-1R1) and IL-1 receptor accessory protein (IL-1RAP), and decreased expression of IL-1 receptor antagonist protein (IL-1RN; the natural inhibitor of IL-1 signalling) by several immune cell subsets. Importantly, microbial compounds are known drivers of IL-1 production¹⁰⁴. Furthermore, patients without GVHD exhibited an upregulation of antimicrobial peptides, such as resistin and defensins on immune effector cells, suggesting antimicrobial efficiency that may have a role in mitigating disease severity.

In a different cohort of patients who had undergone allo-HCT, Han et al.¹⁰³ found significant changes in the gut microbial composition between patients with and without aGVHD, including a reduced abundance of *Bacteroides*, key bacterial producers of SBAs, in the GVHD group and several *Clostridia*-affiliated taxa enriched in the non-GVHD group. Furthermore, those patients with aGVHD had disrupted bile acid metabolism and reduced levels of SBAs including isochenodeoxycholic acid (isoCDCA), 6-keto-lithocholic acid (6-keto-LCA) and 3-oxocholeic acid (3-oxo-CA). In patients without GVHD, these bile acids were positively associated with several enriched *Clostridia* taxa.

Together, the studies outlined in this subsection highlight the role that the intestinal microbiota and its dysbiosis have in, respectively, maintaining or disrupting bile acid homeostasis, as well as the role bile acids have in regulating immune responses after allo-HCT. By identifying mechanistic links between microbiota dysbiosis, bile acid metabolism and immune activation (summarized in Table 1), novel, non-immunosuppressive therapeutic strategies could mitigate aGVHD severity and improve outcomes for recipients of allo-HCT.

It could be concluded that there is a bidirectional relationship between dysbiosis of the gut microbiota and aGVHD onset when tissue injury and immune activation are amplified in a cycle in which GVHD and dysbiosis reinforce one another^{40,68,103}. Although dysbiosis can predispose to GVHD by promoting loss of barrier integrity, reduced production of immunoregulatory metabolites and expansion of pro-inflammatory microorganisms, aGVHD itself could further exacerbate dysbiosis by damaging the intestinal epithelium, increasing oxygen tension and perpetuating inflammation and microbial loss.

Clinical interventions

Although establishing causality from microbiome-targeted interventions in human studies is complex from a biological, clinical, trial design and regulatory standpoint, ongoing trials are investigating several such therapies for clinical practice (Table 2). These efforts are particularly timely and promising, given that currently available non-randomized prospective data demonstrate a clinically meaningful effect size with respect to the association between intestinal microbiota diversity and GVHD-related outcomes. For example, the multicentre study of patients undergoing allo-HCT discussed above²⁴ found that a higher intestinal diversity was associated with a significantly reduced risk of GVHD-related mortality (hazard ratio 0.49; 95% confidence interval 0.26–0.90) over a 24-month follow-up period. This corresponds to an approximate halving of risk and a 7% absolute reduction in

GVHD-related deaths, suggesting a moderate strength of association. Although causality remains unproven, the magnitude of this effect provides a rationale for targeted interventional trials. Such studies will also provide further insight into the biological interplay between hosts and their microbiota that advance efforts to translate observations from mouse models to humans. Current clinical approaches range from non-invasive, such as antibiotic stewardship and dietary modifications, to interventional techniques, such as the administration of probiotics to introduce beneficial bacteria, prebiotics to promote the growth of specific microorganisms, postbiotics to deliver microbial-derived metabolites, and FMT and microbiota-based consortia therapies to reconstitute the microbiota.

Management of antibiotics during allo-HCT

As outlined above, patients undergoing allo-HCT typically receive prophylactic and often therapeutic antibiotics. In addition, some centres administer gut-restricted, non-absorbable antibiotics for gut decontamination with the aim to prevent aGVHD. This practice is based on early preclinical studies suggesting that the eradication of intestinal microorganisms can lessen aGVHD severity^{37,38,105}. However, more recent clinical trials showed no clear advantage^{106–108}, although they were potentially confounded by a lack of treatment standardization across centres, and the most recent randomized phase II trial found no impact on GVHD-related outcomes¹⁰⁹.

Given that broad-spectrum antibiotic exposure in patients receiving allo-HCT is associated with severe perturbations of the gut microbiome and higher incidences and severity of GVHD^{19,22,23,50,110–113}, antibiotic stewardship via evidence-based use of narrow-spectrum antibiotics, ideally for a short duration of time, is prudent to spare beneficial anaerobic bacteria (for example, *Blautia* spp.)^{119,112}. There is also evidence against the use of carbapenems in the first 2 weeks following allo-HCT, which has been associated with higher aGVHD rates in a large retrospective study¹¹⁴. Furthermore, novel approaches used to mitigate antibiotic-induced dysbiosis leading to *C. difficile*-associated diarrhoea may guide the development of microbiome-sparing interventions in patients receiving allo-HCT. For instance, oral administration of either polymer-coated activated charcoal (allowing for its liberation in the ileum and the inactivation of antibiotics in the colon)^{115,116} or β -lactamases, which degrade excess antibiotics, can prevent dysbiosis¹¹⁷, and microbiome-repletion strategies could be used to reintroduce defined microbial consortia¹¹⁸.

Nutritional support during allo-HCT

Several recent studies have explored nutrition-based interventions to mitigate GVHD given the known effects of diet on microbiome composition¹¹⁹. A systematic review and meta-analysis confirmed the superiority of enteral nutrition over parenteral nutrition in reducing the incidence of aGVHD and gastrointestinal aGVHD but found no differences in 100-day overall survival, possibly owing to wide heterogeneity among the analysed studies¹²⁰. Although studies in larger patient cohorts are needed to confirm these results, they are in line with several international guidelines recommending enteral nutrition as the first-line nutritional support for patients undergoing allo-HCT^{121,122}. However, many centres follow traditional clinical practice and primarily use parenteral nutrition. Although there are barriers to the use of enteral nutrition, such as the potential discomfort associated with enteral tube insertion, these recent findings highlight the need to increase awareness about the benefits of enteral nutrition and the overall importance of nutrition for favourable outcomes

Table 1 | Microbiome-derived metabolites associated with allogeneic hematopoietic cell transplantation

Microbial-derived metabolite	Bacterial taxa	Substrates	Mechanisms of action	Therapeutic potential	Select interventions in the USA, all diseases included ^a	Refs.
SCFAs	<i>Faecalibacterium prausnitzii</i> , <i>Bifidobacterium</i> , <i>Ruminococcus</i>	Dietary fibre	Bind to GPCRs, such as GPR43	Anti-inflammatory effects, maintenance of gut barrier integrity	'Mechanisms of Preventing Antibiotic-Associated Diarrhoea and the Role for Probiotics (YOBIOTIC)' (NCT03755765); 'Mechanisms of Probiotics and Antibiotic-Associated Diarrhoea (OURBIOTIC)' (NCT04414722); 'Inulin for Infections in the Intensive Care Unit' (NCT03865706); 'The Influence of Physical Activity on the Gut Microbiome of Pre-Diabetic Adults' (NCT04931836); 'Effect of Spice Consumption on the Microbiome' (NCT03676803)	40
Butyrate or butyric acid	<i>F. prausnitzii</i> , <i>Roseburia hominis</i> , <i>Eubacterium rectale</i>	Dietary fibre, resistant starch	Stimulates mucosal healing; inhibits HDACs	Maintenance of gut barrier integrity and function, immune modulation, such as promotion of T _{reg} cell differentiation, enhancement of microbial diversity, reduced production of pro-inflammatory cytokines, and promotion of anti-inflammatory pathways	'Gut Butyrate and Blood Pressure in African Americans' (NCT04415333)	39–41
Propionate	<i>Bacteroides thetaiotaomicron</i> , <i>Propionibacterium freudenreichii</i>	Dietary fibre	Inhibits pro-inflammatory cytokine release and modulates gut immune environment	Immune regulation, inflammation reduction	'Cardiometabolic Benefits of Potatoes Mediated Along the Gut-Vessel Axis in Adults With Metabolic Syndrome' (NCT03624569)	33,100
Bile acids: taurocholic acid and deoxycholic acid (DCA)	<i>Clostridium</i> spp., <i>Bacteroides</i> spp.	Cholesterol	Interacts with FXR receptors, controlling cytokine production and immune responses; dysbiosis alters bile acid metabolism, promoting inflammation	Regulation of T cell activation and cytokine production	'Ox Bile-Conjugated Bile Acids Sodium in Type II Diabetes Mellitus' (NCT02871882); 'Characterization of Bile Acid Pathway in Obesity' (NCT03341052); 'Thyroid Hormones Homeostasis and Energy Metabolism Changes During Stimulation of Endogenously Secreted Bile Acids' (NCT00706381)	103
Secondary bile acids (SBAs)	<i>Clostridium scindens</i> , <i>Ruminococcus gnavus</i> , <i>Eubacterium</i> spp.	Primary bile acids	Modulates FXR and TGR5 signalling	Inflammation reduction and improved epithelial barrier function	'Effect of Heavy Alcohol Consumption on Farnesoid X Receptor (FXR) Signalling' (NCT02654236)	100,101
Tauroursodeoxycholic acid (TUDCA)	<i>Clostridium</i> , <i>Bacteroides</i>	Primary bile acids	Activates FXR and TGR5 to reduce endoplasmic reticulum stress	Suppression of the release of pro-inflammatory cytokines, enhanced epithelial barrier integrity; FDA approved for liver diseases	'A Trial of Bile Acid Supplementation in Patients With Multiple Sclerosis' (NCT03423121); 'Doxycycline and TUDCA in Patients With Transthyretin Amyloid Cardiomyopathy' (NCT01855360); 'Tauroursodeoxycholic Acid for Protease-inhibitor Associated Insulin Resistance' (NCT01877551)	101,102
Ursodeoxycholic acid (UDCA)	<i>Clostridium</i> cluster XIVa (for example, <i>Clostridium leptum</i> , <i>Clostridium coccoides</i>)	Primary bile acids	Restores bile acid balance	Reduced production of pro-inflammatory cytokines, gut epithelial barrier repair; FDA approved for liver diseases	'Clinical Studies on Bile Acids in Barrett's Oesophagus' (NCT00858858); 'Ursodiol on Insulin Sensitivity, Gastric Emptying and Body Weight With Type 2 Diabetes on Metformin' (NCT02033876); 'C. difficile and Ursodiol' (NCT02748616); 'Phase 2 Study to Evaluate LUM001 in Combination With Ursodeoxycholic Acid in Patients With Primary Biliary Cirrhosis (CLARITY)' (NCT01904058)	102
Lithocholic acid (LCA)	<i>Clostridium</i>	CDCA	Acts through FXR to reduce T cell-driven inflammation and promote immune homeostasis	Anti-inflammatory properties, reduction of pro-inflammatory T cell activity	NA	102

Table 1 (continued) | Microbiome-derived metabolites associated with allogeneic hematopoietic cell transplantation

Microbial-derived metabolite	Bacterial taxa	Substrates	Mechanisms of action	Therapeutic potential	Select interventions in the USA, all diseases included ^a	Refs.
Tryptophan	<i>Bacteroides fragilis</i> , <i>Clostridium sporogenes</i> , <i>Eubacterium hallii</i>	Dietary protein	Converted into bioactive metabolites that regulate immune cells	Restores microbial diversity and enhances production of immune-modulating metabolites; reduces the severity of GVHD	'Mechanistic Approach to Preventing Atrophy and Restoring Function in Older Adults' (NCT02145949)	31
Indole-3-propionic acid (IPA)	<i>C. sporogenes</i>	Dietary protein, tryptophan	Activates the PXR	Antioxidant and anti-inflammatory properties, protection of the gut lining and modulation of immune responses; immune regulation and tolerance, inflammation reduction	'Safety and Pharmacology Study of VP 20629 in Adults With Friedreich's Ataxia' (NCT01898884)	31
Indole-3-aldehyde (IAld)	<i>Lactobacillus reuteri</i> , <i>Lactobacillus plantarum</i>	Dietary protein, tryptophan	Activates the AhR	Promotes IL-22 production by innate lymphoid cells (ILC3s), which is the primary driver of intestinal stem cell regeneration	'Lactobacillus plantarum for GVHD Prevention' (NCT03057054)	91
Isovaleric acid	<i>Bacteroides</i> spp., <i>Prevotella</i> spp.	Branched amino acids	Influences branched-chain amino acid metabolism, modulates immune cell activation	Modulation of systemic branched-chain amino acids homeostasis and attenuation of T _H 1 cell-mediated inflammatory responses	'A study investigating the interaction between gut microbiota and BCAA metabolism to improve metabolic health' (NCT03239717)	40
Lactate	<i>Lactobacillus</i> spp., <i>Bifidobacterium</i> spp.	Carbohydrates	Enhances protein expression of tight junctions in the epithelium and reduces levels of pro-inflammatory cytokines	Maintenance of gut barrier integrity, enhanced epithelial cell survival, and modulation of immune responses to reduce inflammation	'Dietary Supplement With and Without a Probiotic' (NCT04433208)	40
Human milk oligosaccharide (HMO)	<i>Lactobacillus</i> spp., <i>Bifidobacterium</i> spp.	Carbohydrates	Drives the metabolic production of SCFAs and lactate, which lower luminal pH and provide essential energy substrates for colonocytes to maintain intestinal barrier homeostasis	Supports redox homeostasis by favouring the expansion of commensal taxa that mitigate oxidative stress-induced dysbiosis	'Dietary Supplement With and Without a Probiotic' (NCT04433208)	40,79
Urolithins	<i>Akkermansia muciniphila</i> , <i>Eggerthella</i> spp.	Dietary polyphenols	Acts through the AhR, regulating mucosal immunity and gut barrier function	Modulation of mucus production, immune response regulation, maintenance of epithelial barrier integrity; promising for GVHD management	'Pomegranate-Extract Pill in Preventing Tumour Growth in Patients With Localized Prostate Cancer Undergoing Active Surveillance' (NCT02095145)	33
Phenylacetic acid	<i>Clostridium</i> spp., <i>Bacteroides</i> spp.	Phenylalanine	Interacts with T cells and macrophages, influences T _{reg} cell differentiation, modulates cytokine production and immune responses	Reduction of inflammatory signalling in GVHD	'The Impact of the Gut Microbiota on the Metabolism of Branched-chain and Aromatic Amino Acids' (NCT03239717)	103
Methionine-derived metabolites	<i>Bacteroides</i> spp.	Methionine	Regulates redox balance and supports microbial cross-feeding	Microbial resilience, metabolite production; promising for the mitigation of GVHD	'Impact of Gut Microbiota on Methionine and One-Carbon Metabolism' (NCT07283328)	103

This table provides an overview of microbial-derived metabolites and their roles in the development and mitigation of graft-versus-host disease (GVHD) post allogeneic hematopoietic cell transplantation (allo-HCT). It highlights key metabolites, the bacterial taxa that produce them and the substrates involved in their synthesis. In addition, the table lists the mechanisms of action by which these metabolites regulate immune responses, maintain epithelial barrier integrity and suppress inflammation, as well as their therapeutic potential in alleviating GVHD. AhR, aryl hydrocarbon receptor; BCAA, branched chain amino acid; CDCA, chenodeoxycholic acid; FDA, US Food and Drug Administration; FXR, farnesoid X receptor; GPCR, G-protein-coupled receptor; HDAC, histone deacetylase; IL-22, interleukin-22; ILC3, group 3 innate lymphoid cell; NA, not applicable; PXR, pregnane X receptor; SCFA, short-chain fatty acid; TGR5, Takeda G-protein-coupled receptor 5; T_H1 cell, T helper 1 cell; T_{reg} cell, regulatory T cell. ^aThese trials were selected and included to provide proof of concept for the potential use of these interventions in the clinical setting of allo-HCT.

Table 2 | Select, published and ongoing clinical trials investigating microbiota-centred interventions to improve graft-versus-host disease-related outcomes after allogeneic haematopoietic cell transplantation

Intervention type	Status of clinical trial	Clinical trial title	Specific intervention	No. of patients treated	No. of control patients	Clinical trial design	Primary outcome	Results	Phase	Clinical trial identifier	Ref.
Antibiotics	Published	'Microbiota dynamics in a randomized trial of gut decontamination during allogeneic hematopoietic cell transplantation'	Vancomycin–polymyxin B vs no treatment	10	10	Prospective, randomized, open label	Gut microbiome composition	No difference in microbial diversity or GVHD outcomes between treatment and control arm	II	NCT02641236	109
Antibiotics	Active, not recruiting	'Choosing the best antibiotic to protect friendly gut bacteria during the course of stem cell transplant'	Piperacillin–tazobactam vs cefepime for febrile neutropenia	NA	NA	Prospective, randomized, open label	Relative abundance of Clostridiales	NA	II	NCT03078010	170
Microbiota-based consortia therapies	Published	'Treatment of steroid refractory gastrointestinal acute GVHD after allogeneic hematopoietic stem cell transplantation with faecal microbiota transfer (HERACLES)'	Pooled allogeneic faecal microbiota (MaaT013) for the treatment of steroid-refractory gastrointestinal GVHD	24 (HERACLES study)/52 (additional patients from expanded access programme)	NA	Prospective or compassionate use, single arm, open label	Gastrointestinal and overall GVHD response on day 28	Overall response rate 38% (HERACLES)/58% (expanded access programme)	II	NCT03359980	167
Microbiota-based consortia therapies	Active, not recruiting	'MaaT013 as salvage therapy in patients with ruxolitinib refractory gastrointestinal acute GVHD (ARES)'	Pooled allogeneic faecal microbiota (MaaT013) for the treatment of steroid-refractory or ruxolitinib-refractory gastrointestinal GVHD	NA	NA	Prospective, single arm, open label	Gastrointestinal and overall GVHD response on day 29	NA	III	NCT04769895	168
FMT	Published	'Randomized double-blind phase II trial of faecal microbiota transplantation with AML undergoing intensive chemotherapy and patients undergoing transplantation and AML'	FMT in two independent cohorts (patients with AML undergoing intensive chemotherapy and patients undergoing allo-HCT)	26 (AML cohort)/74 (allo-HCT cohort)	NA	Prospective, randomized, double-blind, placebo-controlled	Number of infections until 4 months after the first intervention	No significant reduction in infections	II	NCT03678493	26
FMT	Published	'Donor fecal microbiota transplantation ameliorates intestinal graft-versus-host disease in allogeneic hematopoietic cell transplantation recipients'	FMT in patients with steroid-dependent or refractory gastrointestinal GVHD	15	NA	Prospective, single arm, open label	Gastrointestinal and overall GVHD response up to 6 months after FMT	Complete clinical response in 10 out of 15 patients	I	ISRCTN14530574	154
FMT	Published	'Safety and efficacy of fecal microbiota transplantation for patients with grade IV steroid refractory GI-GVHD: interim results from FMT2017002 trial'	FMT vs no treatment in patients with grade IV steroid-refractory gastrointestinal GVHD	23	18	Prospective, non-randomized, open label	Event-free survival time and overall survival on day 90 after the diagnosis of steroid-refractory gastrointestinal GVHD	Increased median survival time (P=0.002)	I/II	NCT03148743	155
FMT	Active, not recruiting	'FMT for the prevention of acute GVHD in adults undergoing allo-HCT'	FMT in patients undergoing allo-HCT	NA	NA	Prospective, randomized, double-blind, placebo-controlled	Incidence of grade III–IV GVHD up to 6 months after allo-HCT	NA	II	NCT06026371	171

Table 2 (continued) | Select, published and ongoing clinical trials investigating microbiota-centred interventions to improve graft-versus-host disease-related outcomes after allogeneic haematopoietic cell transplantation

Intervention type	Status of clinical trial	Clinical trial title	Specific intervention	No. of patients treated	No. of control patients	Clinical trial design	Primary outcome	Results	Phase	Clinical trial identifier	Ref.
FMT	Recruiting	'Efficacy and safety of autologous FMT in preventing acute GVHD'	Autologous FMT in patients undergoing allo-HCT	NA	NA	Prospective, placebo-controlled, open label	GVHD incidence and severity	NA	I/II	NCT04745221	172
FMT	Recruiting	'FMT in patients with acute gastrointestinal graft-versus-host-disease after allogeneic stem cell transplantation'	FMT in patients with steroid-refractory gastrointestinal GVHD	NA	NA	Prospective, single arm, open label	Sustained remission of gastrointestinal aGVHD	NA	III	NCT03819803	173
FMT	Recruiting	'FMT after allogeneic stem cell transplantation'	FMT vs no treatment after allo-HCT	NA	NA	Prospective, randomized, open label	GVHD incidence and relapse-free survival 1 year after allo-HCT	NA	II	NCT04935684	174
FMT	Active, not recruiting	'Autologous FMT (Auto-FMT) for prophylaxis of <i>Clostridium difficile</i> infection in recipients of allogeneic hematopoietic stem cell transplantation'	Auto-FMT vs no treatment after allo-HCT	NA	NA	Prospective, randomized, open label	<i>C. difficile</i> infection	NA	II	NCT02269150	175
FMT and diet	Recruiting	'FMT and dietary fibre supplementation for the treatment of gut GVHD'	FMT via oral or enteral administration with or without fibre supplementation	NA	NA	Prospective, randomized, open label	Gut microbiome composition, adverse effects	NA	I	NCT05067595	125
Nutrition	Recruiting	'Dietary manipulation of the microbiome-metabolomic axis for mitigating GVHD in patients undergoing allo-HCT'	Starch vs placebo in patients undergoing allo-HCT	NA	NA	Prospective, randomized, single-blind, placebo-controlled	Incidence of grade II–IV GVHD	NA	II	NCT02763033	124
Nutrition	Active, not recruiting	'High dose vitamin A in preventing gastrointestinal GVHD in participants undergoing donor stem cell transplant'	Vitamin A vs no treatment in patients undergoing allo-HCT	NA	NA	Prospective, non-randomized, open label	Adequate vitamin A serum levels without dose-limiting toxicity	NA	I	NCT03719092	176
Nutrition	Active, not recruiting	'Impact of resistance exercise and nutritional endorsement on GVHD symptoms (IRENE-G)'	Supervised resistance training plus nutritional endorsement vs nutritional endorsement only in patients with steroid-treated GVHD	NA	NA	Prospective, randomized, open label	Change in physical performance from baseline to week 24	NA	NA (behavioural intervention)	NCT05111834	177
Prebiotics	Recruiting	'Prebiotic galacto-oligosaccharide (GOS) and acute GVHD'	GOS vs placebo in patients undergoing allo-HCT	NA	NA	Prospective, randomized, double blind, placebo-controlled	Maximal tolerated dose of GOS, incidence of grade I–IV GVHD	NA	I/II	NCT04373057	178
Prebiotics	Completed	'Oral supplementation of 2'-fucosylactose (2FL) in allogeneic bone marrow transplant recipients'	2FL vs placebo in paediatric patients undergoing allo-HCT	NA	NA	Prospective, single arm, open label	Bloodstream infections, number of patients able to take 2FL, change in faecal levels of butyrate, acetate and propionate	NA	I/II	NCT04263597	179

Table 2 (continued) | Select, published and ongoing clinical trials investigating microbiota-centred interventions to improve graft-versus-host disease-related outcomes after allogeneic haematopoietic cell transplantation

Intervention type	Status of clinical trial	Clinical trial title	Specific intervention	No. of patients treated	No. of control patients	Clinical trial design	Primary outcome	Results	Phase	Clinical trial identifier	Ref.
Prebiotics	Recruiting	'Human lysozyme goat milk for the prevention of GVHD in patients with blood cancer undergoing a donor stem cell transplant'	Human lysozyme goat milk (obtained from goats that have been genetically engineered to produce human lysozyme in their milk; human lysozyme is a natural enzyme found in human milk and acts as an antimicrobial) vs no treatment in patients undergoing allo-HCT	NA	NA	Prospective, randomized, open label	Ability to consume human lysozyme goat milk, volume consumed, toxicity, adverse effects	NA	I	NCT04177004	180
Probiotics	Active, not recruiting	'Lactobacillus plantarum in preventing acute GVHD in children undergoing donor stem cell transplant'	<i>L. plantarum</i> vs placebo in paediatric patients undergoing allo-HCT	NA	NA	Prospective, randomized, double blind, placebo-controlled	Incidence of grade I-IV GVHD up to 120 days after allo-HCT	NA	III	NCT03057054	130

Trials were chosen based on methodological rigour, prioritizing those with prospective, randomized and controlled designs when possible. aGVHD, acute graft-versus-host disease; allo-HCT, allogeneic haematopoietic cell transplantation; AML, acute myeloid leukaemia; FMT, faecal microbiota transplantation; GVHD, graft-versus-host disease; NA, not applicable.

following allo-HCT¹²³. Beyond the mode of nutritional intake, several studies are examining nutritional interventions in patients with GVHD (Table 2). For instance, a phase II study at the University of Michigan is assessing the feasibility, safety and potential benefits of modifying the gut microbiome through administering a resistant potato starch supplement to increase intestinal butyrate levels and reduce aGVHD (NCT02763033)¹²⁴. In another example, a phase I study at the Fred Hutchinson Cancer Center is enrolling patients with GVHD to evaluate the safety and feasibility of FMT combined with dietary fibre supplementation (NCT05067595)¹²⁵.

Probiotics and live biotherapeutic products

Probiotics are live microorganisms that can confer a health benefit on the host and are regulated as food or dietary supplements. By contrast, live biotherapeutic products (LBPs) are live organisms designed for targeted therapeutic purposes and subject to clinical drug standards. Both approaches to supplementation provide beneficial microbial strains and offer a potential strategy to restore and reconstitute the disrupted gastrointestinal microbiota for patients with aGVHD. Although infrequent, probiotics have been described to cause bloodstream infections in patients who are immunocompromised¹²⁶, so testing for safety and efficacy is crucial.

Several approaches have been tested. For example, the administration of *Lactobacillus* alleviates GVHD in mouse models^{127,128}, but a clinical trial examining *Lactobacillus rhamnosus* GG administration in patients receiving allo-HCT was terminated early when an interim analysis did not detect a probiotic-related change in the microbiome or in GVHD incidence²⁷. *Lactobacillus plantarum* is under investigation in paediatric patients receiving allo-HCT. Its safety has been confirmed¹²⁹, and a randomized phase III trial is underway to assess its efficacy in the prevention of aGVHD in this patient population (NCT03057054)¹³⁰. *Clostridium butyricum* MIYAIRI 588 (CBM588) is being studied as an LBP in a randomized open-label pilot study with peri-transplant administration. Patients in the treatment group showed increased alpha diversity in their gut microbiome and lower abundance of *Enterococcus* and *Bacteroides*, which were both more common in patients who developed GVHD. However, the sample size of 11 treated patients was not large enough for outcomes analysis¹³¹. SER-155 is an investigational oral LBP being studied to improve clinical outcomes in patients receiving allo-HCT. Although the bacterial species in SER-155 are proprietary, its composition has been rationally designed based on taxa associated with reduced risk of bloodstream infections and GVHD in human studies. Initial results presented at the 2024 Tandem Meetings of American Society for Transplantation and Cellular Therapy (ASTCT) and Center for International Blood and Marrow Transplant Research (CIBMTR) demonstrated that patients tolerated the product well, drug strains successfully engrafted and patients had reduced pathogen domination in their gut microbiomes¹³². Results from a randomized, double-blinded, placebo-controlled phase I study of SER-155 are awaited (NCT04995653)¹³³.

Prebiotics

Several prebiotic therapeutic approaches for mitigating GVHD have been tested in recent years. Fructo-oligosaccharides (FOS) are well-studied prebiotic carbohydrates that have successfully been used in patients with inflammatory bowel disease¹³⁴. After studies in mouse models found that FOS supplementation increased intestinal SCFA levels¹³⁵, which in turn can lower GVHD severity, a phase I dose-escalation trial demonstrated that FOS were well tolerated in

recipients of allo-HCT, supporting the feasibility of supplementation for modulation of the gut microbiota¹³⁶. Additionally, a pilot randomized clinical trial of the commercially available probiotic supplement FamiLact (comprising seven beneficial bacterial strains) in combination with FOS found a reduced risk of severe aGVHD and increased levels of T_{reg} cells in a small intervention group undergoing allo-HCT¹³⁷. Building on single-centre retrospective and preclinical studies indicating a beneficial effect of glutamine, fibre and oligosaccharide (GFO) and resistant starch supplementation^{138–140}, a non-randomized prospective study of these supplements demonstrated significantly reduced GVHD severity in patients receiving allo-HCT relative to a historical control group¹⁴¹. GFO and resistant starch may promote beneficial gut bacteria, particularly *Bifidobacterium*, increasing the production of SCFAs and enhancing integrity of the gut barrier. Furthermore, a recent pilot study found administration of resistant starch alone to be feasible and well tolerated in patients receiving allo-HCT, and that the intervention may mitigate disruption of the microbiome and associated metabolites¹⁴². A phase II trial is now ongoing (NCT02763033)¹²⁴. Results from another randomized, multicentre, phase IIa study investigating the safety and efficacy of orally administered *B. infantis* combined with human milk oligosaccharides in patients undergoing allo-HCT are awaited (NCT06102213), but it should be noted that the sponsor made the business decision to terminate the trial early¹⁴³. The secondary outcome measures of the trial include incidence of bloodstream infections, neutropenic fever and aGVHD. Several other clinical trials investigating various prebiotic strategies are also ongoing (Table 2).

Postbiotics

Postbiotic strategies involve the direct administration of beneficial metabolites or molecules of bacterial origin, thus reducing the risk of infection compared with live probiotics. This could make postbiotics a safer therapeutic alternative, particularly in patients who are immunocompromised receiving allo-HCT. As discussed above, preclinical studies have demonstrated beneficial effects of several postbiotics in mouse models of GVHD, such as the alleviation of GVHD severity after restoration of butyrate or propionate levels^{52,53}. Although no clinical studies examining the effects of SCFA administration during allo-HCT have yet been published, these postbiotics have shown beneficial immunomodulatory effects in gastrointestinal inflammatory disorders^{144,145}. Other promising approaches based on findings from mouse studies are the supplementation of xylose (a *B. ovatus* product inhibiting mucus-degrading bacteria that compromise the intestinal barrier)⁴² and SBAs^{100,101,103}, but clinical trials are needed to evaluate these hypotheses.

Faecal microbiota transplantation

FMT, involving the introduction of processed stool from a donor (often a healthy volunteer) into the gastrointestinal tract of a recipient, is perhaps the best-studied microbiota-centred intervention for preventing and treating GVHD. First used to treat severe *C. difficile* infections, an FMT product is approved by the US Food and Drug Administration (FDA) for this purpose^{146,147}. It is also a promising approach to prevent dysbiosis or to reconstitute a perturbed gut microbiome in patients undergoing allo-HCT and has been used to treat refractory *C. difficile*-associated diarrhoea and to decolonize the gut of multidrug-resistant bacteria in these patients^{148–150}. However, challenges with respect to safety and standardization must be addressed to ensure clinical efficacy of FMT for patients receiving allo-HCT, particularly because they are severely immunocompromised and at high risk

for bacterial translocation from FMTs. Indeed, a documented transmission of a multidrug-resistant *Escherichia coli* from a single FMT donor to two recipients of allo-HCT resulted in one death, underscoring the need for strict donor screening¹⁵¹. Nevertheless, multiple pilot studies have shown that, in most cases, administration of either autologous or healthy donor faecal material to patients after allo-HCT is feasible and safe, and can restore microbial diversity while reducing the abundance of potentially pathogenic taxa^{152,153}.

Early studies of FMT lacked standardization of patient and treatment variables. Historically, FMT has often relied on loosely regulated, heterogeneous preparations. Combined with inconsistent criteria for recipient selection, this variability may help explain the range of outcomes reported across studies. Furthermore, a substantial proportion of human data to date come from retrospective or non-randomized prospective designs, which limits causal interpretation. Nonetheless, several single-arm studies have indicated potential for clinical benefit for some steroid-refractory or high-risk, therapy-naïve patients with aGVHD^{154–162}. Two studies that included a control group also provide some evidence of a benefit from treatment^{163,164}, but reasons for response or non-response in recipients of FMT remain largely elusive. Notably, the first randomized, double-blind, placebo-controlled study of third-party FMT administered at the time of neutrophil recovery following allo-HCT found a generally favourable impact on the composition of the microbiome, but no improvement in clinical outcomes. Furthermore, the 4-month all-cause infection rate, which was the primary end-point, did not improve, and the study even reported a numerically higher incidence of grade III–IV aGVHD after FMT, although differences in prophylactic regimens for GVHD likely confounded these results²⁶. A subset analysis by the researchers revealed variability in the engraftment of donor FMT-derived bacterial strains and identified low microbial diversity pre-FMT as a predictor of engraftment, suggesting that patients with lower diversity might particularly benefit from FMT¹⁶⁵. Another recent study indicated that donor choice can be critical for FMT engraftment and clinical outcomes¹⁶⁶. Therefore, a paradigm shift is likely needed: FMT must become a rigorously developed, standardized and regulated therapeutic modality, akin to any other pharmaceutical product. This includes adherence to Good Manufacturing Practice standards, stringent donor screening, quality control and regulatory oversight. Such a transition is essential not only to ensure product consistency and patient safety but also to allow for the integration of FMT into standard medical care.

Microbiota-based consortia therapies

Microbiota-based consortia therapies represent an advancement in the field of microbiota interventions: products such as MaaT013 are derived from pooled donors and standardized to capture a broader and more diverse microbial ecosystem. Their composition is more complex than LBPs, aiming to replicate the functional resilience of a healthy microbiome rather than targeting specific pathways or diseases. The clinical relevance is exemplified by the HERACLES trial, a prospective, international, single-arm, phase II study of the safety and efficacy of MaaT013 (ref. 167). Among 24 patients with steroid-resistant gastrointestinal aGVHD taking part in the trial, the gastrointestinal overall response rate (defined as complete resolution of gastrointestinal aGVHD symptoms) was 38%, whereas an additional 52 patients treated under a French expanded access programme showed an overall response rate of 58%. Responders in both cohorts exhibited increased microbiota diversity and higher abundances of beneficial bacteria, particularly producers of butyrate, along with elevated levels of SCFAs

Glossary

Alloreactive donor T cells

Donor-derived T lymphocytes that recognize recipient human leukocyte antigen (HLA) as non-self and initiate an immune response.

Alpha diversity

A measure of microbial diversity within a single sample, capturing both the number of taxa present (richness) and their relative distribution (evenness). Commonly quantified using indices such as the Shannon index or Simpson index.

Bacteraemia

The presence of bacteria in the bloodstream.

Bacteriophages

Viruses that infect and replicate within bacteria, influencing bacterial populations and gene transfer within the gut microbiome.

Bayesian modelling

A statistical approach that uses prior knowledge and probability to interpret complex data, commonly used to model biological systems such as immune responses.

Bronchiolitis obliterans syndrome

A chronic lung condition involving inflammation and scarring of the small airways, which can occur as a complication after allogeneic haematopoietic cell transplantation.

Cholestatic diseases

A group of disorders marked by impaired bile formation or flow, leading to cholestasis (reduced or obstructed bile secretion). These conditions may be intrahepatic or extrahepatic and often progress to liver injury.

Endogenous pyrogens

Fever-producing factors that originate from within the human body, often in response to inflammation.

Enteral nutrition

Administration of nutrients to the gastrointestinal tract by oral intake or via a tube.

Enterotypes

Distinct clusters within the gut microbiota characterized by the dominance of certain bacterial genera, used to categorize individual microbiome profiles.

Facultative anaerobes

Microorganisms that can grow in both the presence and absence of oxygen, allowing them to adapt to varying oxygen levels in the gut. By contrast, strict anaerobic microorganisms cannot grow in the presence of oxygen as they lack enzymes to neutralize harmful oxygen by-products.

Faecal microbiota transplantation

(FMT). The process of transferring stool from a healthy donor into the gastrointestinal tract of a patient to restore balance to the microbiome.

Genetic convergence

The independent emergence of similar genetic traits or functions across distinct microbial taxa or communities, often driven by shared environmental pressures or host interactions.

Human leukocyte antigen

(HLA). A group of highly polymorphic cell surface proteins that are part of the HLA system. These molecules are the human equivalent of the major histocompatibility complex (MHC) molecules found in other species. HLA proteins present peptide antigens to T cells, having a critical role in immune recognition and response.

Indoles

A class of aromatic heterocyclic compounds derived from the amino acid tryptophan. Indoles are produced by bacterial metabolism and act as signalling molecules that can influence host immunity, intestinal barrier function and microbial ecology.

Major histocompatibility complex (MHC) class II

Cell surface glycoproteins expressed primarily on professional antigen-presenting cells and that present extracellular-derived peptide antigens to CD4⁺ T helper cells, initiating and regulating adaptive immune responses.

Mucositis

Painful inflammation and ulceration of the mucus membranes lining the digestive tract, frequently caused by chemotherapy or radiation.

Mycobiome

The community of fungi that reside in and on the human body.

Myeloablative regimens

Treatment regimens involving high-dose chemotherapy or radiation to destroy bone marrow cells, typically used before stem cell transplantation.

Neutropenia

A low number of neutrophils (a type of white blood cell) in the blood, commonly occurring after chemotherapy and associated with a high risk of infection. Febrile neutropenia is also associated with fever.

Non-myeloablative regimens

Less intensive pre-transplant regimens that do not completely eliminate the recipient's bone marrow, aiming to reduce toxicity while allowing donor cell engraftment.

Parenteral nutrition

Intravenous administration of nutrients.

Postbiotics

Non-living microbial products or metabolites that confer health benefits to the host without containing viable organisms.

Prebiotics

Dietary substances, typically fibre, that promote the growth and activity of beneficial gut bacteria.

Probiotics

Live microorganisms that, when administered in sufficient amounts, confer a health benefit on the host.

Secondary bile acids

(SBAs). Bile acids formed in the intestine by microbial modification of primary bile acids through deconjugation and dehydroxylation. SBAs modulate lipid absorption in the host.

Short-chain fatty acids

(SCFAs). Microbial fermentation products, such as butyrate, propionate and acetate, which support gut health and regulate immune responses.

Third-party FMT

Faecal microbiota transplantation (FMT) where the faecal matter has been sourced from a donor distinct from the patient's transplant donor.

Tight junction

A specialized intercellular junction that seals adjacent epithelial or endothelial cells, forming a selective barrier that regulates paracellular transport of ions, solutes and water. Tight junctions maintain tissue integrity and contribute to barrier function.

Vancomycin-resistant *Enterococcus*

(VRE). A strain of *Enterococcus* bacteria resistant to the antibiotic vancomycin, capable of causing serious infections, particularly in patients who are immunocompromised.

Virome

The collection of all viruses, including bacteriophages and eukaryotic viruses, that inhabit the human body.

and SBAs by day 28. By contrast, non-responders had higher levels of pathogenic pro-inflammatory bacteria and increased levels of systemic inflammatory markers. However, severe infectious complications, including sepsis, were observed in five patients. Bacteria from the donor product were not detected in any of these cases, leaving the connection to the intervention uncertain. Furthermore, the prospective cohort showed lower overall survival (28% at 6 months) compared with both the expanded access programme group (49%) and survival estimates obtained from an analysis of 25 studies on second-line treatment for steroid-refractory aGVHD (49% weighted average 6-month overall survival)¹⁶⁷. This discrepancy is likely due to the HERACLES trial enrolling a particularly high-risk population with severe gastrointestinal involvement and high-risk prognostic scores, whereas the expanded access programme included some patients with less severe disease and may have benefited from delayed effects of prior treatments such as ruxolitinib. Following these encouraging results, a phase III, single-arm trial was conducted across 50 sites in Europe (ARES (NCT04769895)¹⁶⁸), evaluating the efficacy of MaaT013 in 66 patients with steroid-refractory and ruxolitinib-refractory gastrointestinal aGVHD. The findings have not yet been peer reviewed but, according to an abstract presented at the 2025 annual meeting of the American Society of Hematology, the primary end-point of the study was met, with a gastrointestinal overall response rate at day 28 of 62%. Furthermore, 12-month overall survival was significantly higher in responders versus non-responders (67% versus 28%, $P < 0.0001$)¹⁶⁹.

Conclusion and future perspectives

Substantial progress has been made in recent years towards understanding the role of the intestinal microbiota in outcomes following allo-HCT, and in particular its influence on GVHD pathogenesis, metabolic homeostasis and immune regulation. The intestinal metabolome has emerged as a central mediator in this process, with distinct metabolic shifts linked to GVHD development. Together, these findings offer promising avenues for novel therapeutic interventions that target the microbiome and its metabolic outputs. Although there is growing enthusiasm for such microbiota-targeted strategies, the current body of evidence remains complex and, at times, contradictory. Although several studies have reported associations between microbial diversity, targeted interventions and aGVHD-related outcomes, not all findings have been consistent. These discrepancies highlight the need for a more nuanced understanding of causality and effect size, as well as the clinical relevance of microbiota modulation.

Nonetheless, the field is advancing rapidly. Emerging interventions such as FMT and microbiota-based consortia therapies show promise, although their clinical application remains in the early stages. Importantly, several clinical trials are underway and expected to report results that will elucidate the therapeutic potential of these approaches. Future multicentre studies incorporating advanced metagenomic and metabolomic analyses will also be essential to prospectively characterize microbiota dynamics and metabolite profiles during allo-HCT and GVHD. In parallel, mechanistic studies are needed to further elucidate how the microbiome influences immune responses and tissue homeostasis. These efforts will pave the way for the development of predictive biomarkers and personalized, microbiota-modulating therapies. Although challenges remain, we are optimistic that microbiota-centred interventions will ultimately become an integral part of clinical practice in allo-HCT.

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Competing interests

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